A Maturity-Governed Rotational Support Hypothesis for Disk Galaxies: Preliminary Tests and Falsifiable Predictions

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Abstract

Established input. SPARC supplies baryonic masses, observed flat rotation velocities, gas fractions, surface-density proxies, and H I extent measures for a quality-controlled disk-galaxy sample; some galaxies occupy different locations relative to the mature BTFR scale. Decisive same-object resolved H I morphology and kinematic measurements are not yet available for a sufficiently complete homogeneous sample, so the present work is a preliminary scalar diagnostic rather than a confirmation of the mechanism. **Preliminary result.** We ask whether a maturity-dependent BDVR governor hypothesis can order the normalized support residual A_{inferred} using only f_*, Σ_b, f_{gas}, and $R_{\text{H I}}/R_d$. The selected product gate gives modest ordering with substantial unexplained scatter: MSE=0.183 and Spearman $\rho = 0.146$. The result is further limited because the clipped target has 12 galaxies at 0, 6 at 1, and only 69 galaxies in the open interval. **Unproven prediction.** Resolved H I morphology, velocity-field symmetry, velocity dispersion, and settling measurements should distinguish low-activation from mature-response systems. External surveys are therefore used as proxy context and candidate-selection channels, not as tests or validation of the governor.

1 Introduction

The empirical regularity of disk-galaxy rotation remains a central clue in galaxy astrophysics. The baryonic Tully-Fisher relation (BTFR) and the radial acceleration relation (RAR) show that baryons and rotational response are tightly organized, while SPARC makes it possible to inspect how flat velocities relate to gas fraction, surface density, morphology, and H I extent. The question pursued here is narrower than a full galaxy-dynamics theory: can a maturity-dependent governor hypothesis generate a falsifiable ordering of disk-galaxy rotational support?

We adopt a bound-domain vacuum response (BDVR) formalism in which galaxy-scale rotational support includes an effective response that may activate with baryonic organization. The physical hypothesis is that a dynamically organized bound baryonic disk may receive additional effective rotational support, and that the response may depend on structural maturity. The empirical diagnostic in this paper is deliberately weaker: bounded logistic factors built from f_*, Σ_b, f_{gas}, and $R_{\text{H I}}/R_d$ define a phenomenological test surface.

Operational definition. Dynamically immature or low-organization state denotes a gas-rich, weakly converted, low-activation state. The shorthand juvenile is used sparingly and does not require youth in cosmic time.

This paper does not claim that the governor hypothesis has been confirmed. The available scalar data permit a preliminary test of whether maturity-related variables can order the inferred rotational response, but they do not provide the resolved same-object H I and kinematic measurements required to establish the proposed mechanism. The principal result is therefore a falsifiable observational specification: the hypothesis predicts that low inferred activation should coincide with gas-rich, extended, asymmetric, or dynamically unsettled disks, while mature activation should coincide with coherent rotational support. The required measurements are identified here as the next test rather than assumed to exist.

2 Physical Hypothesis and Claim Boundary

The physical hypothesis is not a validation claim: a dynamically organized bound baryonic disk may receive additional effective rotational support, and the response may depend on structural maturity. The empirical diagnostic is the bounded product gate below. It is a phenomenological test surface, not derived from an action or fundamental field equation.

3 SPARC Sample and Measured Quantities

The SPARC diagnostic sample supplies $v_{\rm obs}$, baryonic mass components, disk scale lengths, gas fractions, surface-brightness proxies, and H I extent where available. We use $r_{\rm eval}$ as the SPARC flat-velocity/evaluation radius associated with the tabulated outer velocity diagnostic. The baryonic mass convention is $M_{\rm bar} = M_* + 1.33 M_{\rm H\,I}$; H I masses are multiplied by 1.33 to account for helium. Stellar masses use the SPARC 3.6 micron stellar mass-to-light ratio $\Upsilon_{3.6} = 0.5$, and bulge mass is included when present in the SPARC baryonic mass.

4 Velocity Normalization and Inferred Activation

We write the flat-velocity governor equation as

$$v_{\rm flat}^2 = v_N^2 + A_{\rm inferred} Q_{\rm eff} \sqrt{G M_{\rm bar} a_0}, \qquad 0 \le A_{\rm inferred} \le 1. \tag{1}$$

Here v_N is the Newtonian baryonic circular speed at $r_{\rm eval}$, $v_{\rm obs}$ is the observed SPARC flat velocity, $v_{\rm full}$ is the full-response BDVR velocity scale, and $a_0 = 1.2 \times 10^{-10}\,{\rm m\,s^{-2}}$ is the acceleration scale used in the BTFR-like response term. For this scalar diagnostic we set $Q_{\rm eff} = 1$ as a normalization convention for all galaxies, so the full-response velocity scale is

$$v_{\rm full}^2 = v_N^2 + \sqrt{G M_{\rm bar} a_0}. \tag{2}$$

A physical structural-response model would have to specify $Q_{\rm eff}$ independently before Equation (1) could be predictive. In this paper, $Q_{\rm eff}$ is not fitted per galaxy and is not used to tune $A_{\rm inferred}$. The analysis uses three related but distinct quantities:

1. $\alpha_{\rm BTFR}$ is purely empirical: the observed response relative to the BTFR scale,

$$\alpha_{\rm BTFR} = \left(\frac{v_{\rm obs}^4}{G M_{\rm bar} a_0} \right)^{1/2}. \tag{3}$$

2. $A_{\rm inferred}$ is the observed support requirement, a normalized residual under the assumed BDVR velocity decomposition, renormalized by v_N^2 and clipped to $[0,1]$:

$$A_{\rm inferred} = {\rm clip}_{[0,1]} \left[\frac{v_{\rm obs}^2 - v_N^2}{v_{\rm full}^2 - v_N^2} \right]. \tag{4}$$

When $v_{\rm obs}^2 < v_N^2$, $A_{\rm inferred}$ is set to 0; when $v_{\rm obs}^2 > v_{\rm full}^2$, it is set to 1.

3. A_{pred} is the model prediction based on maturity gates. It asks how much support the governor hypothesis predicts from f_*, Σ_b, f_{gas}, and $R_{\mathrm{H\,I}}/R_d$, without using the observed rotation endpoint.

5 Phenomenological Maturity-Gate Model

The logistic function used by each gate is

$$\sigma(x; x_c, k) = \frac{1}{1 + \exp[-k(x - x_c)]}. \tag{5}$$

The product-gate model is

$$A_{\mathrm{pred}} = A_* A_\Sigma A_{\mathrm{gas}} A_{\mathrm{H\,I}}, \tag{6}$$

with $A_* = \sigma(f_*; f_{*,c}, k_*)$, $A_\Sigma = \sigma(\log \Sigma_b; \log \Sigma_{b,c}, k_\Sigma)$, $A_{\mathrm{gas}} = 1 - \sigma(f_{\mathrm{gas}}; f_{\mathrm{gas,c}}, k_{\mathrm{gas}})$, and $A_{\mathrm{H\,I}} = 1 - \sigma(R_{\mathrm{H\,I}}/R_d; (R_{\mathrm{H\,I}}/R_d)_c, k_{\mathrm{H\,I}})$. Stellar conversion and surface-density buildup open the gate; high gas fraction and extended H I suppress it.

6 Preliminary SPARC Results

Across the current SPARC diagnostic sample, maturity-class predicted-activation summaries are: juvenile: N=12, median A_{pred}=0.042; mature: N=36, median A_{pred}=0.555; transition: N=39, median A_{pred}=0.349. The single-variable correlations with A_{inferred} are modest, which is expected if activation is not controlled by one maturity coordinate alone. The useful separation appears only when the gates are combined. In the product model, juvenile systems are strongly suppressed, transition systems sit near the middle of the switch, and mature systems move into the higher-response channel. The diagnostic result is not "stellar fraction alone controls the response." The result is that gas-rich, weakly converted, H I-extended systems fail the combined maturity gate, while compact and more settled disks are allowed into the mature rotational-support channel.

Table 1: Excluded asymmetric scalar case retained as a follow-up target.

Galaxy	f_*	f_{gas}	$R_{\mathrm{H\,I}}/R_d$	α_{BTFR}	A_{inferred}	A_{pred}
UGC07125	0.180	0.820	6.82	0.389	0.000	0.057

UGC07125 is excluded from clean scalar model tests after the published geometry audit. Its scalar coordinates are reported for transparency: $f_* = 0.180$, $f_{\mathrm{gas}} = 0.820$, $R_{\mathrm{H\,I}}/R_d = 6.82$, $\alpha_{\mathrm{BTFR}} = 0.389$, $A_{\mathrm{inferred}} = 0.000$, and $A_{\mathrm{pred}} = 0.057$. Published observations indicate side-dependent H I and stellar geometry: the W side is warped, while the E side is not warped; DDO161 is the cleaner juvenile-forming contrast.

7 Robustness and Limitations

The threshold grid scans stellar-fraction, surface-density, gas-fraction, and H I-extent breakpoints with fixed logistic slopes. Each grid point is scored against A_{inferred}, not against the descriptive maturity class. The score combines mean-squared prediction error with diagnostic penalties that require low predicted activation for juvenile systems and high predicted activation for a substantial mature subset. These penalties do not replace the fit to A_{inferred}; they reject activation surfaces

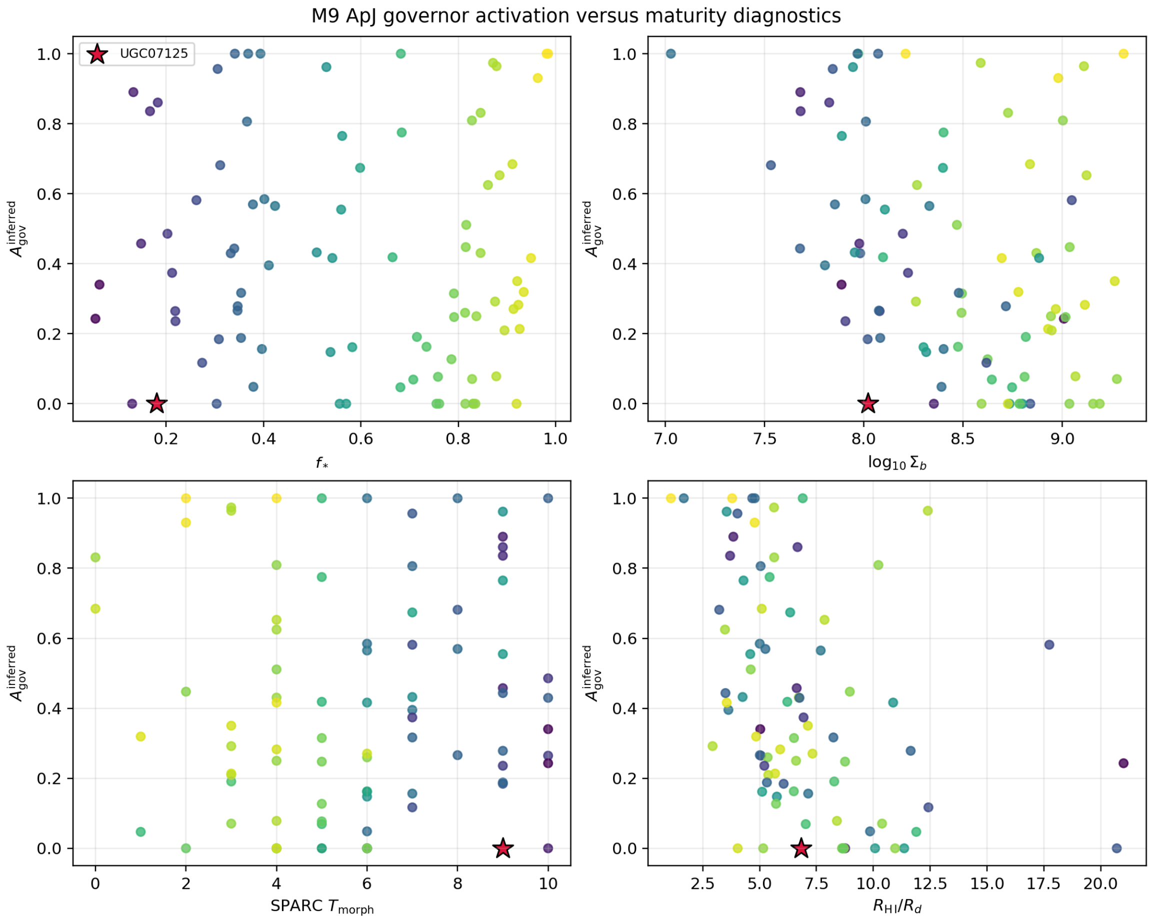

Figure 1: Marginal maturity diagnostics for the SPARC diagnostic sample (N=87). **Clipping warning:** inferred activation has strong boundary concentrations; see Section 7 for the numerical clipping audit. These panels are marginal diagnostics rather than independent tests, and the boundary concentrations should not be over-interpreted. Spearman correlations are rho(fstar)=-0.014, rho(fgas)=0.014, rho(log Sigma b)=-0.364, rho(T)=0.082, and rho(RHI/Rd)=-0.500.

that fit numerically only by allowing physically wrong behavior, such as sending gas-rich juvenile systems through the mature channel.

Table 2: Selected product-gate maturation model. The MSE and ρ ranges summarize the ten highest-ranked grid points, so the table reports the selected surface without repeating near-identical rows.

Rank	Variant	$f_{*,c}$	$\log \Sigma_{b,c}$	$f_{\rm gas,c}$	$(R_{\rm H\,I}/R_d)_c$	Slopes	MSE range	ρ range
best	product_star_sigma_gas_hi	0.25	7.80	0.75	7.5	$k_* = 10$, $k_\Sigma = 4$, $k_{\rm gas} = 8$, $k_{\rm H\,I} = 1.0$	0.183–0.199	0.075–0.146

The selected surface has $f_{*,c} = 0.25$, $\log \Sigma_{b,c} = 7.8$, $f_{\rm gas,c} = 0.75$, and $(R_{\rm H\,I}/R_d)_c = 7.5$. Its diagnostics are MSE=0.183 and Spearman $\rho = 0.146$; this is modest ordering with substantial unexplained scatter, not validation or confirmation. Details of the grid search are in Appendix A.

Clipping audit. The clipped $A_{\rm inferred}$ distribution contains 12 galaxies clipped to 0, 6 galaxies clipped to 1, and 69 remaining within the interval. Thus 18 of 87 galaxies are boundary-coded rather

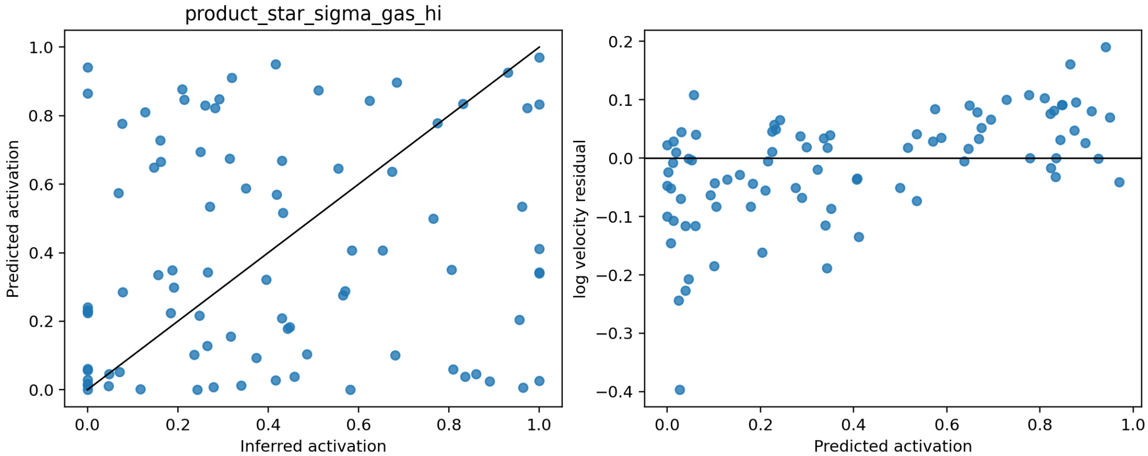

Figure 2: Predicted versus inferred activation. This is the core preliminary diagnostic: predicted activation is computed from maturity gates, while inferred activation is computed from the observed velocity endpoint. The identity line and baseline comparisons should be read against the broad scatter; the current scalar product gate provides only modest predictive ordering and does not constitute confirmation of the governor hypothesis.

than continuously inferred from the data. Only 69 of 87 galaxies remain in the open interval, so A_{inferred} is not a reliable continuous regression target in the current normalization. The unclipped activation proxy spans -0.19 to 1.57 with median 0.34; therefore the vertical concentrations at 0 and 1 are partly consequences of the normalization and clipping rule. Sensitivity of the fit to clipping remains a central limitation.

Table 3: Baseline comparison for preliminary SPARC activation ordering.

Model	MSE	Spearman ρ	Note
constant mean baseline	0.104	nan	no maturity variables
stellar fraction alone	0.255	-0.018	f_* only
gas fraction alone	0.255	-0.018	$1 - f_{\text{gas}}$ only
surface density alone	0.291	-0.366	$\log \Sigma_b$ only
H I extent alone	0.157	0.497	$R_{\text{H I}}/R_d$ suppressor only
reduced two-variable model	0.248	-0.018	f_* and f_{gas}
selected product gate	0.183	0.146	four logistic gates
simple linear regression	0.084	0.490	linear fit to four maturity variables

The selected product gate has higher MSE than the constant-mean baseline in Table 3, so MSE should not be read as evidence for a successful continuous regression. Because the endpoint target is heavily clipped, low/high activation classification is the more appropriate first use of the diagnostic. The four-gate form is retained only as a hypothesis-generating ordering device; the marginal improvement over stellar fraction or gas fraction alone must be tested in an independent resolved-observation sample.

Resampling and sensitivity. A repeated k-fold cross-validation diagnostic using fixed predictions gives held-out prediction error MSE $= 0.183 \pm 0.048$ across folds. A bootstrap confidence

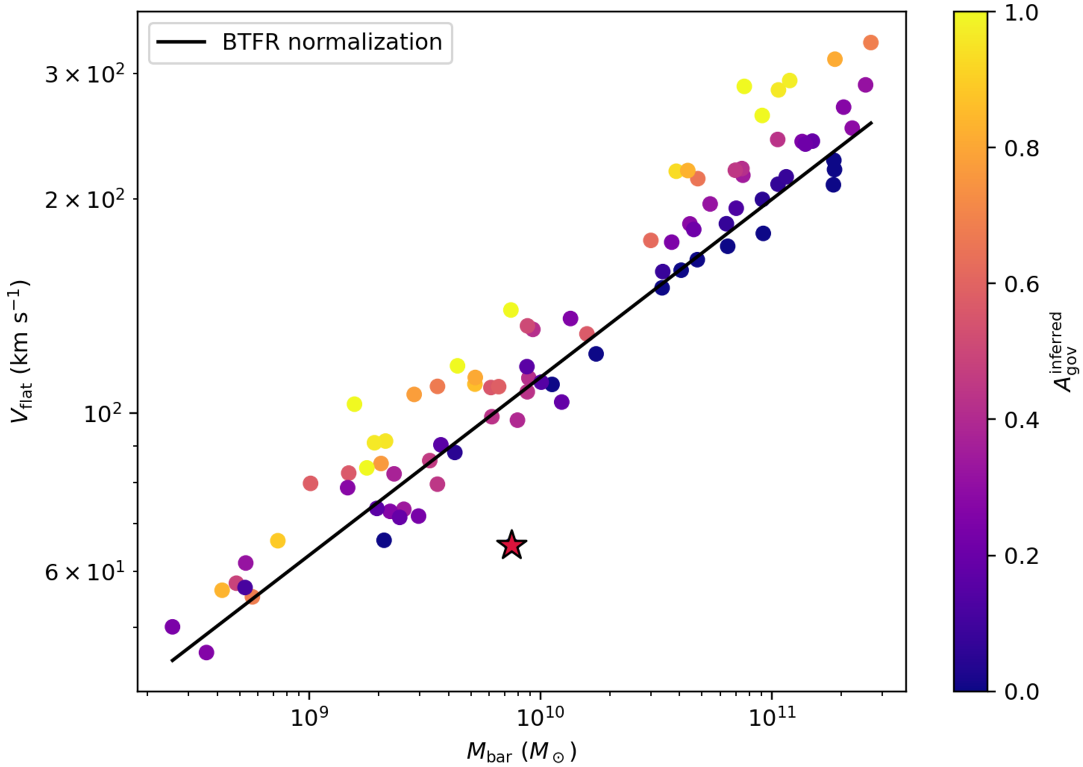

Figure 3: BTFR plane colored by inferred governor activation. The coloring is descriptive because A_{inferred} is partly constructed from the same observed velocity on the vertical axis; it is not an independent validation plot.

interval for Spearman ρ is approximately -0.094–0.311; threshold stability across resamples is only a preliminary audit because the current grid was selected on the same sample. Inclusion/exclusion sensitivity for the known asymmetric geometry case gives MSE=0.183 with the case included and MSE=0.185 for the case excluded. asymmetric/warped-system exclusions and alternate quality cuts are required before any stronger claim. Full uncertainty propagation remains deferred; distances, inclinations, v_{flat}, stellar mass-to-light ratio, gas mass, disk scale length, H I radius, and evaluation radius must be propagated before galaxies near $A_{\mathrm{inferred}} = 0$ are interpreted as physically distinct from low positive values.

8 Corrected Activation-Rank Diagnostic

The corrected activation-rank calculation is included as a supervised reparameterization diagnostic. It asks whether the same corrected scalar target can be ordered by a latent coordinate after expanding clipped endpoints and adding maturity, reservoir, and angular-momentum context. The baseline product gate has Spearman $\rho = 0.146$ against corrected A_{inferred}. The selected corrected rank coordinate reaches full-fit rank Spearman $\rho = 0.981$ and five-fold out-of-fold rank Spearman $\rho = 0.968$; the out-of-fold bounded RMSE is 0.081.

This improvement is manuscript-relevant because it shows that the corrected endpoint target contains an orderable scalar structure. It is not independent validation: the coordinate is trained

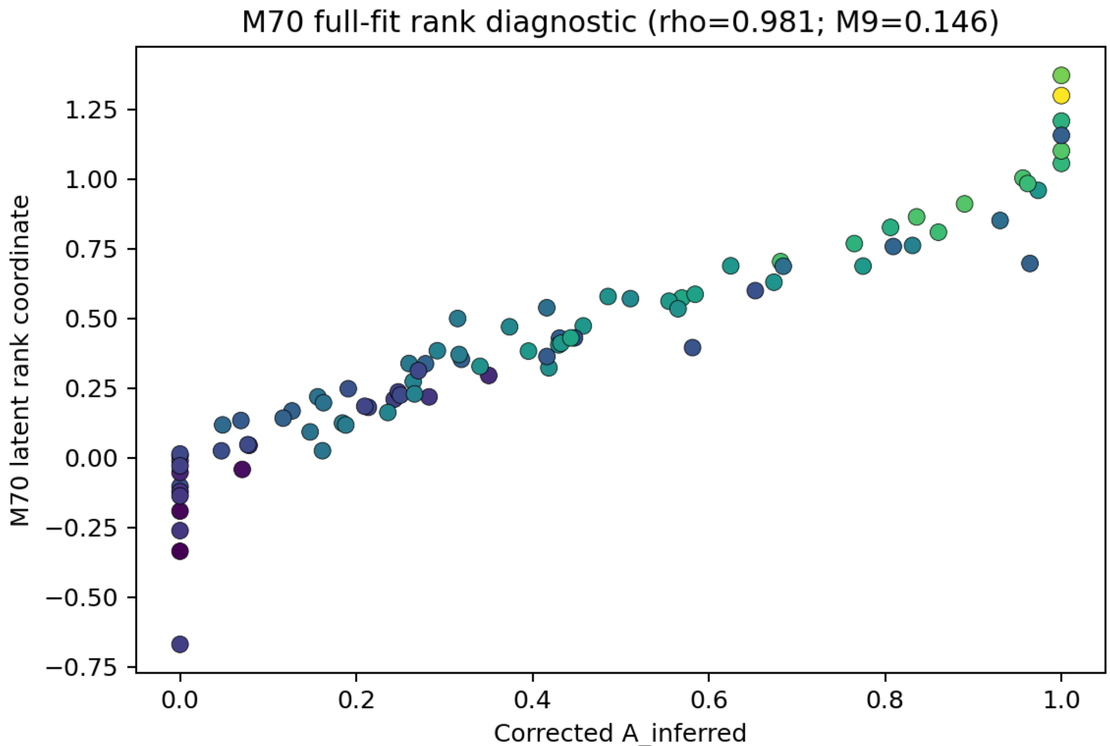

Figure 4: Corrected activation-rank reparameterization diagnostic. The panels compare corrected inferred activation with the supervised latent rank coordinate and its bounded version. The figure is a diagnostic of orderability within the current scalar target, not independent validation of the physical governor.

on corrected A_{inferred}, and its success does not promote the governor, does not change the mature velocity law, and does not replace the required same-object H I morphology and kinematic test. The selected rank coordinate uses angular-momentum context; M59 backbone context is not used by the selected model, so proxy/context rows are not promoted to direct validation. UGC07125 remains excluded from clean scalar interpretation, with $A_{\text{inferred}} = 0.000$ and bounded corrected-rank value $A_{\text{bounded,M70}} = 0.000$.

9 A Testable Maturation Sequence

The juvenile-to-mature hypothesis predicts that gas consumption and disk settling raise the activation state. A single-zone toy track captures the direction of the observable sequence: a gas-rich protogalaxy begins with low f_* and low A_{pred}, then approaches the mature BTFR as star formation and structural settling increase the activation gate.

10 External Proxy Consistency Checks and Target Selection

External surveys provide proxy consistency checks and candidate-selection channels, not direct validation of BDVR activation. THINGS is the most relevant pilot because it contains resolved H I maps, but N=8 is exploratory and motivates a larger homogeneous audit. MaNGA supplies stellar

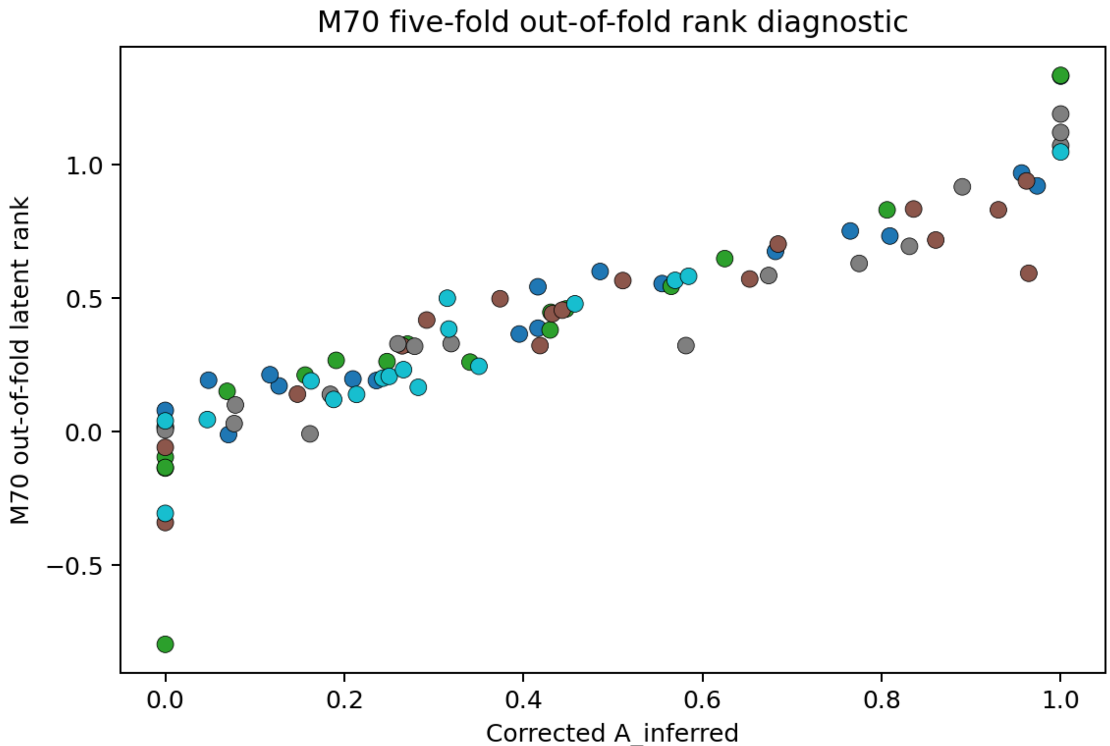

Figure 5: Five-fold out-of-fold corrected-rank diagnostic. The out-of-fold comparison guards against a purely in-sample display, but it remains trained on the same corrected activation target and should be read as a reparameterization audit rather than an external validation test.

and ionized-gas IFU proxies, not outer H I rotation or direct governor activation. xGASS/xCOLD GASS is used for candidate selection and gas-state ordering. ATLAS3D fast and slow rotators are endpoint controls, not direct disk-galaxy analogs or evidence for a BDVR maturation sequence.

Table 4: External survey context for target selection; these rows are not BDVR activation tests.

Dataset	Context	Qualitative use	Direct-test limitation
THINGS	Low-A_{pred} galaxies should have higher H I disturbance.	N=8 resolved-H I pilot; useful for selecting an asymmetry follow-up sample.	Too small for a statistically meaningful BDVR activation test.
MaNGA	Dense, low-sSFR systems occupy an earlier-type/settled locus.	Large IFU proxy sample for target triage and inner-settling context.	MaNGA supplies stellar and ionized-gas IFU proxies, not outer H I rotation.
ATLAS3D	Fast and slow rotators bracket mixed-support and pressure-supported endpoints.	Endpoint controls for language about rotational support channels.	Early-type fast/slow rotators are not disk-galaxy activation tests.
xGASS/xCOLD GASS	Gas fraction and depletion track ordinary gas-richness/maturity trends.	Candidate selection and gas-state ordering for future resolved follow-up.	Gas-fraction correlations are partly definitional, not independent BDVR evidence.

This table is a target-selection map, not a hypothesis-test table. External surveys show qualitative trends consistent with the idea that gas-rich systems are less dynamically settled, but they do not provide direct tests of the BDVR activation hypothesis. As context, the ladder still separates gas-rich diffuse systems, settling disks, mixed-support fast rotators, and pressure-supported spheroids into qualitatively different response channels. THINGS has only an N=8 pilot overlap,

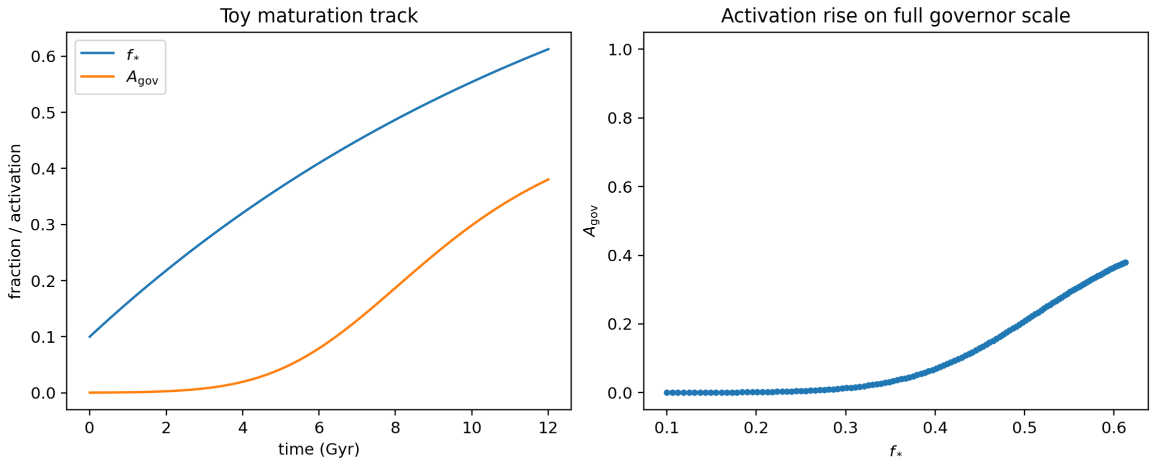

Figure 6: Conceptual toy trajectory, not a reconstructed evolutionary history. The track illustrates a hypothesized direction in maturity space. Any assumed evolution of baryonic mass, disk scale length, and surface density is a toy prescription, not a source-derived history.

MaNGA uses an invented scalar proxy rather than outer H I rotation, xGASS/xCOLD GASS gas-fraction correlations are partly definitional, and ATLAS3D fast/slow rotators are endpoint controls rather than disk-galaxy activation tests.

11 Candidate Targets

The current direct audit records 2 low-activation candidate(s); DDO161 is retained as the cleaner juvenile-forming contrast. The broader ladder ranks 8 THINGS resolved-H I proxy rows, 80 xGASS/xCOLD GASS gas-maturity rows, 80 MaNGA scalar/IFU proxy rows, and ATLAS3D control endpoints in a unified 70-row synthesis table. The 70-row synthesis table reports candidate pools and controls after dataset-specific ranking; every candidate still requires geometry or outer-curve verification before direct scalar interpretation.

Table 5: Direct SPARC low-activation audit.

Galaxy	A_{inferred}	A_{pred}	f_*	f_{gas}	$R_{\mathrm{H\,I}}/R_d$	BTFR offset	Status
DDO161	0.000	0.013	0.130	0.870	8.76	-0.060	clean juvenile-forming contrast; resolved H I follow-up still required
UGC07125	0.000	0.057	0.180	0.820	6.82	-0.205	excluded from clean scalar test: side-dependent H I/stellar geometry; W side warped, E side not warped

Table 6: Analog-search channels.

Channel	Dataset	Role	N searched	Candidate count	Next action
SPARC direct audit	SPARC	direct	87	2	resolved H I and velocity-field check
Resolved H I	THINGS	proxy	8	8	homogeneous H I asymmetry audit
Gas maturity	xGASS/xCOLD GASS	proxy	80	80	rotation-curve and H I follow-up
Scalar/IFU screen	MaNGA	proxy	80	80	gas and outer-curve completion
Endpoint control	ATLAS3D	control	80	80	contrast pressure-supported channel

Table 7: Top low-activation follow-up candidates and juvenile-forming contrasts.

Candidate	Dataset	Score	Has H I map	Has IFU	Priority
DDO161	SPARC	0.871	N	N	Clean juvenile-forming contrast
12700-12701	MaNGA	0.836	N	Y	High—follow-up target
9091-6103	MaNGA	0.829	N	Y	High—follow-up target
GASS102001	xGASS/xCOLD GASS	0.827	N	N	High—follow-up target
UGC07125	SPARC	0.823	N	N	Excluded from clean scalar test—side-dependent H I/stellar geometry
8611-6104	MaNGA	0.822	N	Y	High—follow-up target
GASS108119	xGASS/xCOLD GASS	0.792	N	N	Medium—proxy screen
8139-3703	MaNGA	0.785	N	Y	Medium—proxy screen
8617-12705	MaNGA	0.781	N	Y	Medium—proxy screen
8619-12704	MaNGA	0.778	N	Y	Medium—proxy screen

The tables separate immediate activation checks from follow-up targets and excluded boundary objects. Direct SPARC rows already have activation, baryon, H I-extent, and BTFR coordinates, but published geometry determines whether those scalar coordinates are clean. External candidates are ranked follow-up targets: gas-selected rows need outer rotation and resolved H I information, IFU rows need gas and outer-curve completion, and endpoint/control rows test the boundary of the maturation sequence rather than direct disk activation.

12 Observational Predictions and Falsification Program

Headline falsification test. The strongest prospective test is resolved H I asymmetry and settling: low-A_{pred} galaxies selected before observation should show systematically larger H I asymmetry, stronger approaching/receding mismatch, larger centroid offsets, larger warp or position-angle variation, or higher turbulent/dispersion support than mass-matched high-A_{pred} controls. The current SPARC thresholds are retrospective and are not independent falsification tests until fixed before a new sample is measured.

Predictions already testable with SPARC concern scalar locations in A_{pred}, A_{inferred}, and BTFR response. Predictions requiring resolved H I concern asymmetry, warp or position-angle variation, approaching/receding rotation-curve mismatch, and centroid offsets. Predictions requiring IFU plus outer rotation curves concern velocity dispersion and inner settling. High-redshift predictions require different tracers and calibrations and should be treated as prospective tests rather than retrospective thresholds. The decisive new dataset is approximately 20 low-A_{pred} targets and approximately 20 mass-matched high-A_{pred} controls with resolved H I maps, approaching/receding rotation curves, velocity-dispersion estimates, H I asymmetry, warp and position-angle variation, stellar/H I centroid offsets, and consistent distances and inclinations. No existing public table currently combines all of these measurements homogeneously for the required objects.

1. **Prediction 1 (mature systems).** Among galaxies with $f_* > 0.3$, $\log \Sigma_b > 8.0$, $f_{\text{gas}} < 0.5$, and $R_{\text{H I}}/R_d < 5.0$, less than 5% should have $A_{\text{inferred}} < 0.2$.

2. **Prediction 2 (juvenile systems).** Among galaxies with $f_* < 0.2$ and $R_{\text{H I}}/R_d > 6.0$, less than 10% should have $A_{\text{inferred}} > 0.6$.

3. **Prediction 3 (H I asymmetry).** In a sample of 20 low-A_{pred} galaxies and 20 matched high-A_{pred} controls, the median H I asymmetry should be at least 0.1 higher in the low-activation sample, with a Mann–Whitney $p < 0.05$.

4. **Prediction 4 (high-redshift gas-rich disks).** In a mass-matched sample at $z > 1$, the median $\alpha_{\text{BTFR}} < 0.8$ for galaxies with $f_{\text{gas}} > 0.7$.

The governor hypothesis fails if, after controlling for baryonic mass and measurement quality, low predicted activation is not associated with reduced rotational response or with independently measured dynamical immaturity. It also fails if dynamically settled, compact, stellar-rich disks frequently occupy the low-response channel. Thresholds above are retrospective when selected from the current SPARC sample and prospective only when fixed before a new resolved-observation campaign.

13 Discussion: Relation to MOND, the RAR, and Disk Evolution

The acceleration scale and BTFR-like term have established precedent in Milgromian dynamics and RAR work. This paper does not claim that BDVR independently rediscovers ordinary scaling relations. The RAR gas-richness trend already shows that gas-rich systems occupy a recognizable sequence in acceleration space; the proposed difference is the threshold-and-settling interpretation, namely that resolved disk organization should determine which gas-rich, extended systems remain in a low-response channel. The current scalar evidence is too weak to distinguish that interpretation from a standard gas-richness trend. A reader could reasonably ask why not use stellar fraction or gas fraction alone: Table 3 shows that those one-variable proxies have similar rank information, and the four-gate model offers only marginal improvement. The product gate is therefore a target-selection hypothesis, not a demonstrated new scaling law. In ordinary ΛCDM hydrodynamic simulations, disk settling also correlates with gas fraction, surface density, and velocity dispersion; Toomre-style stability language is useful only as motivation for future resolved disk-settling variables, not as the definition of Q_{eff} here. The distinguishing observation is whether same-object resolved H I asymmetry, velocity-field mismatch, and settling measurements predict the normalized support residual after controlling for standard mass, size, and measurement-quality variables.

14 Conclusions

The present analysis motivates, but does not establish, the hypothesis that any additional BDVR-like rotational response is not universal across baryonic disks. The scalar SPARC diagnostics provide only modest evidence for an ordering in which gas-rich and H I-extended systems occupy a lower predicted-response channel; the severe clipping limitation in Section 7 makes this a low/high activation screen, not a continuous physical activation measurement. The decisive test requires resolved same-object H I morphology and kinematics that are not presently available in a sufficiently complete homogeneous sample. The next paper-worthy result is therefore observational: approximately 20 low-A_{pred} targets, approximately 20 mass-matched high-A_{pred} controls, resolved H I maps, side-specific rotation curves, dispersion estimates, asymmetry measures, warp/position-angle variation, centroid offsets, and consistent distances and inclinations. If that dataset does not link low predicted activation to reduced response or independently measured dynamical immaturity, the governor hypothesis fails.

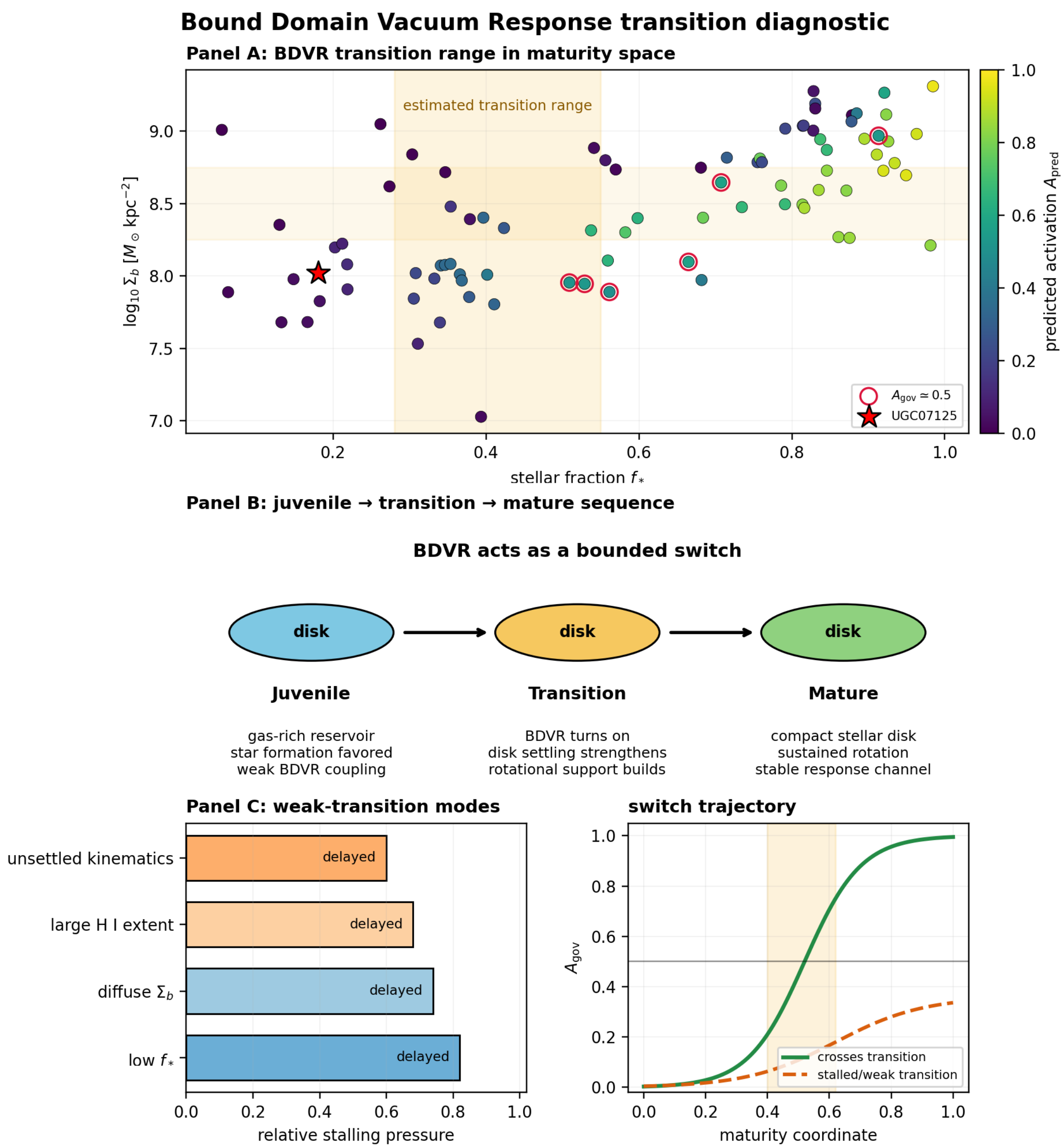

Figure 7: Multi-panel BDVR transition diagnostic. Panel A marks the maturity-space transition range where stellar conversion and surface-density buildup begin to open the response channel. Panel B walks through the juvenile, transition, and mature sequence. Panel C shows weak-transition modes that leave gas-rich or diffuse systems below sustained mature rotational support.

A Threshold grid-search details

The grid search is a diagnostic selection layer, not a fundamental equation. It scans a compact set of stellar-fraction, surface-density, gas-fraction, and H I-extent breakpoints with fixed logistic slopes, scores each surface against A_{inferred}, and applies diagnostic penalties only to exclude numerically adequate surfaces with physically wrong channel behavior. The selected product gate is then carried into candidate selection and external proxy consistency checks.

B Test Results Summary

The SPARC quality-1 governor run contains 87 galaxies. The toy maturation diagnostic classifies 14 dynamically immature systems, 32 transition systems, and 41 mature systems. The selected threshold surface has mean squared error 0.183 and Spearman $\rho = 0.146$. The external proxy synthesis contributes 11275 usable proxy measurements across five survey stages, with the external sequence ordered by gas fraction, surface density, angular-momentum support, and pressure support.

C Dataset Descriptions

SPARC supplies the quality-1 disk-galaxy flat velocities and baryonic variables used for direct governor activation inference. THINGS supplies resolved H I structure maps for pilot disturbance and side-to-side kinematic diagnostics. MaNGA supplies a large IFU scalar sample used for surface-density and star-formation proxy activation. ATLAS3D supplies early-type angular-momentum measurements used for fast-rotator bridge and pressure-channel diagnostics. xGASS and xCOLD GASS supply atomic and molecular gas measurements used for gas-depletion maturity ordering.

D Machine-Readable Artifacts

The manuscript package includes machine-readable source tables copied beside the manuscript for reproducibility:

- `m9_summary.json`
- `m9_inferred_activation.csv`
- `m9_governor_threshold_fits.csv`
- `m9_governor_predictions.csv`
- `m9_gate_diagnostics.csv`
- `m9_ugc07125_governor_audit.csv`
- `m9_falsification_candidates.csv`
- `m16_external_validation_summary_table.csv`
- `m16_external_validation_claim_boundaries.csv`
- `m16_external_validation_synthesis.json`

- m17_ugc07125_prototype_vector.json
- m17_sparc_direct_analog_audit.csv
- m17_ugc07125_analog_synthesis.csv
- m17_ugc07125_analog_synthesis.json
- m70_summary.json
- m70_predictions.csv
- m70_cv_predictions.csv
- m70_reparameterization_fits.csv
- m70_fold_metrics.csv
- m70_equation_specification.md

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